

Visual Question Answering on 3D Scenes

Shubham M Mahale

Dept. of Computer Science and Engineering
PES University
Bengaluru, India
shubhammahale49@gmail.com

Y Nikhil Bharadwaj

Dept. of Computer Science and Engineering
PES University
Bengaluru, India
nikhil.yellapragada@gmail.com

Yamajala Siddhardha

Dept. of Computer Science and Engineering
PES University
Bengaluru, India
yamajalasiddhardha1@gmail.com

Altaf A Kandagal

Dept. of Computer Science and Engineering
PES University
Bengaluru, India
altafkandagal01@gmail.com

Surabhi Narayan

Dept. of Computer Science and Engineering
PES University
Bengaluru, India
surabhinarayan@pes.edu

Abstract— AI agents are the ones which perceive the environment and take action accordingly to achieve the goal. The main task of the AI agent is to mimic the behavior of humans and perceive the environment similar to what humans do. When we hear the word "Agent perceives an environment" the first thing that comes to our mind is observing the environment and identifying objects present in the surroundings. This brings us to our problem statement that is "Visual Question answering on 3D scenes". The problem setting consists of multiple scenes wherein each scene is described by 4 images. Images capture the different views of the geometrical objects placed randomly in the scene. The purpose of this paper is to give the agent a 3-dimensional view of the environment so that no details regarding the objects in the environment are hidden. So, that the agent can perceive the environment and ascertain the positions of various objects in the environment relative to each other. This helps the agent to understand the environment more accurately as compared to a 2-D image wherein many details are concealed due to the fixed field of view resulting in a critical lack of details. The aim of this paper is to enhance the current world of robotic technology as it can be applied in human robot interaction systems and real-life robots and help them understand and perceive the environment. We use modern ML, NLP approaches to solve the problem.

I. INTRODUCTION

It's quite natural for humans to perceive the world around them using their eyes, ears and other senses. Humans share their thoughts to other people through either a common spoken language or written language. However for an AI agent to perform complex tasks, it is important to perceive its environment. Computer vision(CV) and Natural language processing(NLP) [16,8,6] are the fields which enables the AI agent to perceive the environment and share its thoughts through a natural language mimicking the human behaviour. "Visual Question Answering"(VQA) which deals with machine answering any question asked by the user on a image has received attention from various AI communities and was termed as an "AI-complete" task involving both CV(Computer Vision) and NLP(Natural Language Processing) techniques as it can help in advancement of the AI world by leveraging this technique in real life intelligent robots so that the daily human tasks can get simpler.

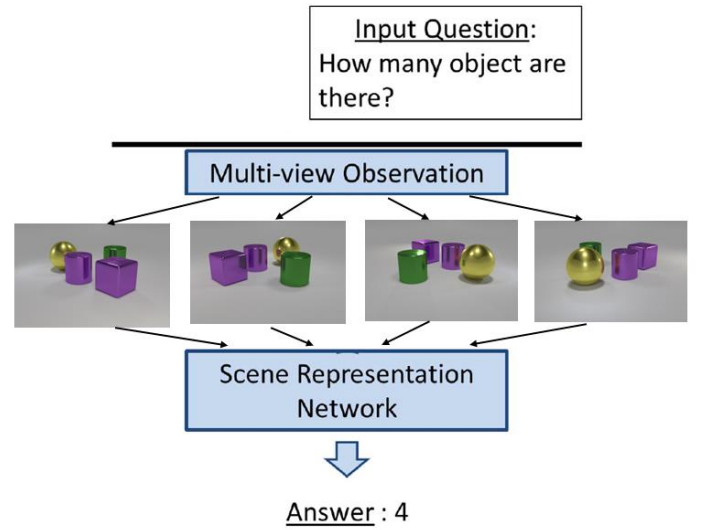


Figure1: Illustration of proposed multi-view Visual Question Answering.

The most recent works reported in the literature on VQA focuses on answering questions based on a single RGB images. However, in real-world environments, relying solely on visual information from a single RGB image may not be sufficient due to factors such as poor lighting, inappropriate viewpoint and occlusion. Additionally, for various computer vision (CV) and natural language processing (NLP) multi-modal tasks, recognizing 3D information is crucial. For example, in human-robot interaction (HRI) systems [21], as humans perceive and recognize their surroundings based on the 3D information [2], it is essential for robots to establish recognition in a similar way with human operators for better communication. Hence we focus on leveraging traditional visual question answering into "Visual Question answering on 3D scenes" i.e. we take multiple views of the image known as viewpoints and create a 3D view of a image. The dataset used is CLEVR dataset containing 3D rendered images in JSON format and contains different questions and answers about the scene. Considering the hardware limitations, execution time, The use of dataset with direct 3D information was avoided.

The paper is organised into 5 sections. Section II describes the related works in VQA, section III is the

problem statement, section IV is the proposed methodology, section V is results and discussion and section VI describes conclusion.

II. RELATED WORKS

A. Visual Question Answering:

In real-life scenarios, single viewpoint images may not provide sufficient visual information for accurate question answering due to factors such as lack of visibility and inappropriate viewpoint. Therefore, incorporating 3D information [2] is essential for a variety of computer vision and natural language processing tasks. There are methods to bridge the gap between 3D and language information in the context of VQA.

Typically, in VQA models that works on 2d images will have images represented by features obtained from pretrained CNNs. For representing questions, word embeddings or recurrent networks are used. The recurrent networks may be trained for generating answers or for multiclass classifications over common answers, or for binary classifications on image-question-answer triples. Many methods include attention over the image or question, while some methods involve memory or dynamic network architecture.

LSTM Q + I (Long Short-term Memory question+image) extracts image and question features using CNN and LSTM separately, and fuses them by point-wise multiplication operation, and then predicts answer from fused features through fully-connected layer and SoftMax function. Most current VQA methods also use this base structure. Since multi-modal features representation is one of the crucial components of the VQA method, the authors of proposed an efficient multi-modal compact bilinear pooling (MCB) [9] as an alternative to the outer product calculation of multi-modal features.

The VQA 2.0 [24] dataset is widely used and includes genuine images from the MS COCO dataset as well as questions-answer pairs collected through crowdsourcing on Amazon Mechanical Turk. However, Johnson et al. [18] pointed out that the VQA 2.0 [24] dataset suffers from significant biases in dataset distribution. To address this issue, they introduced the CLEVR [8] (Compositional Language and Elementary Visual Reasoning) dataset and a dataset generation process that effectively controls these biases. In our research we make modifications to the CELVR dataset setup to generate multi-vies images and regulate the rate of occlusion.

B. Embodied Question Answering (EQA):

The EQA task, as defined by Das et al. [11], involves an agent placed within a 3D environment, which aims to answer questions by gathering image information through navigation based on egocentric perceptions. To accomplish this task, the authors propose a baseline approach that divides the problem into a recurrent neural network (RNN) based hierarchical planner-controller navigation policy and a VQA model which combines the weighted sum of the image and question features to produce an answer. Other studies have expanded the EQA task to include more complex scenarios, such as multiple targets and photorealistic environment with point cloud data. Most EQA methods separate vision language grounded navigation process from

the VQA process because the former process is more important. While most EQA methods treat navigation and VQA separately, our work concentrates on integrating 3D and language information to improve VQA performance.

C. 3D Representation

Recently a series of works apply CNN (Convolutional Neural Network) to point cloud data for 3D representation learning. There are two critical issues in applying conventional CNN to point cloud data: the order lessness of point cloud data and the invariance to rigid transformations. PointNet [12] uses a max pooling-based symmetric function to handle the order lessness problem and introduces a spatial transformer network to maintain the invariance to the rigid transformations.

There have been numerous instances of 3D representation learning from multi-view images [13]. Multi-view images representation can efficiently use 2D CNNs, making them perform well with lower processing time and memory cost than other data forms like 3D voxel data or point cloud data. In a study, a multi-view CNN (MVCNN) structure was suggested for object recognition from multi-view images. In this method, the multi-view images representation is achieved by adding a view pooling operation after separately extracting image features from multi-view. Another study, RotationNet [14], further expands the MVCNN structure by simultaneously predicting object categories and 6D poses. Eslami et al [22]. proposed Generative Query Network (GQN) to jointly train a scene representation from multi-view images and a rendered to produce an image from a given viewpoint and scene representation. We introduce GQN to learn scene representation [22] and apply this representation to VQA tasks. Our model can directly obtain a compact 3D scene representation from multiple viewpoints, making our approach valuable. Additionally, SPLATNet proposed in [13] makes it possible to fuse 2D image information with 3D point cloud by introducing a bilateral convolution layer structure which can deal with data with different dimension.

III. PROBLEM STATEMENT

Most recent works in VQA deal with a 2-Dimensional RGB image. However, for real-life applications the information obtained from the view perspective of the RGB images becomes insufficient for a task such as visual question answering due to this may be due to various factors that influence the quality of the image such as poor lighting conditions while the image is captured, inappropriate viewpoint, occlusion etc. This can lead to the machine not understanding and inferring the environment correctly. Thus, leading to an incorrect result. To avoid this, we incorporate 3D view to perform the task of visual question answering. Using 3D information to recognize the objects is an important task in Computer Vision (CV) and NLP [16,8,6] tasks. Recognition based on 3D Scenes is vital because we as humans also perceive and recognize the environment around using a 3-Dimensional view, it is essential for the agent to do the same to enhance the communication between the human interacting with it. So, our problem statement is to integrate 3D information obtained from multiple viewpoints of a single image and perform visual question answering.

It consists of 3 modules:

1) *Question Featurization*: Converting the questions in natural language into their natural embeddings for further processing. Identifying the features of the objects present in the images for further processing.

2) *Object detection*: Identifying the features of the objects present in the images for further processing.

3) *Joint Feature Representations*: Combining the features of the objects and the questions to help the model identify the objects and answer the question asked.

IV. PROPOSED METHODOLOGY

This paper outlines our approach for utilizing the rendered CLEVR dataset [8] and ask relevant questions related to the objects in the environment to the model to predict the answer. Each scene contains different types of geometric objects like cube, cylinder, sphere that have properties like texture, color and size present at random positions in a scene. Each scene will have 4 view-points images to which questions can be asked.

A. Overall Framework

Traditional VQA methods predict the answer from a given question and a single image. In real environment applications, recognition based on 3D information [2] is necessary to answer a given question in various situations, such as scenes with severe occlusion. Therefore, we propose combining 3D information with the question and answer information, especially, considering the memory and execution time costs, we try to capture the 3D information from Multiview images of the given scene.

The CLEVR dataset [8] that contains 3D rendered images in JSON format and blender is used to render multi-view images in a scene. After the generation of multiple views, we will be using an R-CNN [23] to process the image and a GRU [20] to process the text part of the inputs. These will then be sent through a joint feature representation module and the output text will be generated.

B. Dataset Generation

The dataset that we are using is called Compositional Language and Elementary Visual Reasoning (CLEVR) [8] is a synthetic visual question answering dataset. It contains 3D-rendered images containing objects, each image comes with a number of related questions. Each object that is rendered contains random attributes like size, color, material, shape and are placed randomly in a plane. We have generated CLEVR dataset images with multiple views and have also provided the user the freedom to change the number of viewpoints, number scenes and the resolution of the image generated. We have fixed the number of view points to 4. We set the virtual cameras (viewpoints) around the scene every 90 degrees, and these cameras are at an elevation of 45 degrees from the scene plane. All cameras are set to look at the centre of the scene.

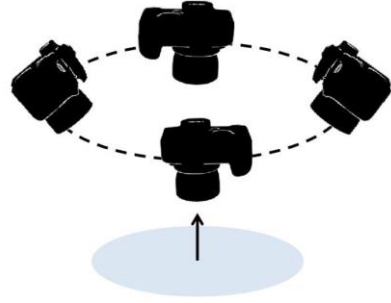
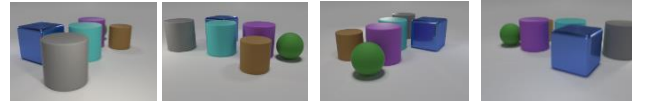


Figure 2: Illustration of the cameras placed around the scene spaced every 90 degrees and elevated 45 degrees from the bottom scene plane.



Viewpoint 1 Viewpoint 2 Viewpoint 3 Viewpoint 4

Figure 3: Multi-viewpoint images

In addition, we modified the object attribute setups present in the dataset generation program so that, in our dataset, each scene has 3 to 5 objects with different colors (like yellow, red, blue, purple, and green), sizes (large and small), material (metallic and matte), shapes (cube, cylinder, and sphere), and different location distributions. Each scene is rendered at the end.

We basically use Blender 2.79 version to render the multi-view images. In order to render each type of scene, base blender files are used. The base blender files contain raw rendered images of each type of objects or shapes. Then the dataset generation code uses this blender files as input and adds the attributes like size, color, texture to it. Similarly based on the number of objects selected for a scene, each scene is rendered at the end.

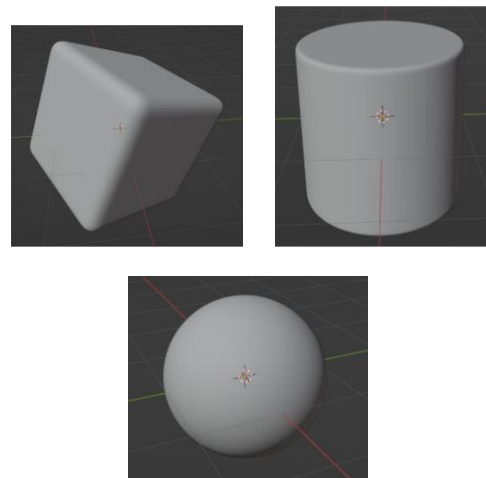


Figure 4: Base Blender mask images

Since we are working on multi-view dataset and we need 4 view-point images of each scene, some properties and number of shapes selected are used to render all four view-points in a scene.

The dataset generation part also includes setting of proper resolution of images rendered and changing the number of scenes and view-point per scenes. Images are rendered in batches according to the number of scenes and view-points per images. We can also change minimum and maximum number of images per scene.

Once the images are rendered, all the images are stored in an output file with proper naming convention of the file names. Along with that for each scene separate JSON files are also created which contains all the properties of the objects in the scene. Till now we have generated around 700 scenes, each scene will have 4 viewpoint images so in total we have rendered around 2800 CLEVR images. This becomes our dataset for the VQA.

C. Question Generation

CLEVR questions are generated using the script `generate_questions.py`. This script reads JSON file containing information about scenes and outputs a JSON file containing questions, functional programs and answers for those images. Questions are generated using question templates.

For each scene 10 unique questions with answers are generated and are put in the JSON files along with the image names. These values present in the JSON files along with the images are then used to do further Visual Question Answering

The methodology is divided into 2 main phases:

1) Image Feature Extraction

2) Text feature extraction + Image features + attention map

1) *Image Feature Extraction*: This is the first phase where we have performed image feature extraction using DenseNet121 [25] architecture of CNN. We have a backup of the features extracted so that they can be used later. The main reason for choosing this architecture is that it solves the problem of vanishing gradient and mainly has lesser number of parameters to train, hence this approach was used

2) *Text Feature Extraction + Joint Feature Representation*: In the second phase we performed text preprocessing on the questions using methods such as lemmatization and tokenization to make sure that questions are uniformly phrased.

We then used Attention maps to focus/extract the main features or we can say the required features of both image and question and generated the corresponding context vectors, we did the following process with word level features + image features wherein we first find the attention maps of both and pass these maps to the context vector

function to generate the context vectors and pass it through a Dense layer to get the final vector.

Similarly, we repeat the process for phrase level features + image features and sentence level features + image features (Here image features correspond to the extracted image features from the DenseNet121). We do this process in a hierarchical process i.e., by appending the previous level features to the next level and use a classifier at the final layer which is again the Dense layer to get the final answer to the asked question, we used the embedding module of keras to represent the word level features, for phrase level features we used a user defined function and finally for sentence level feature extraction we made use of a LSTM

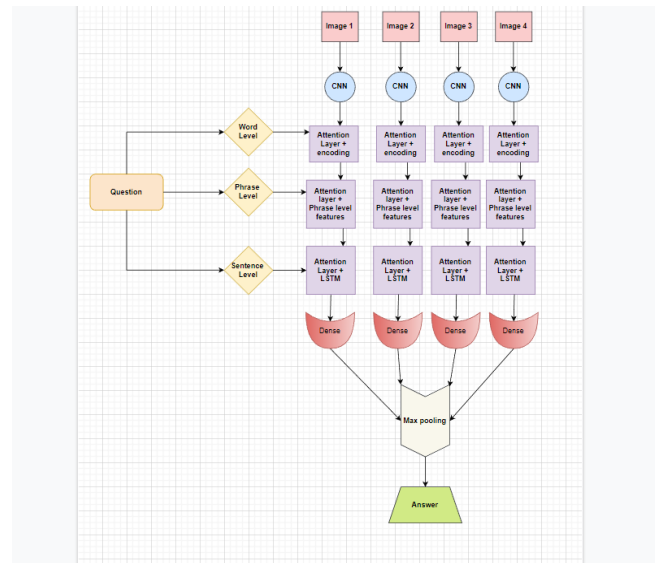


Figure 5: Hierarchy of Architecture

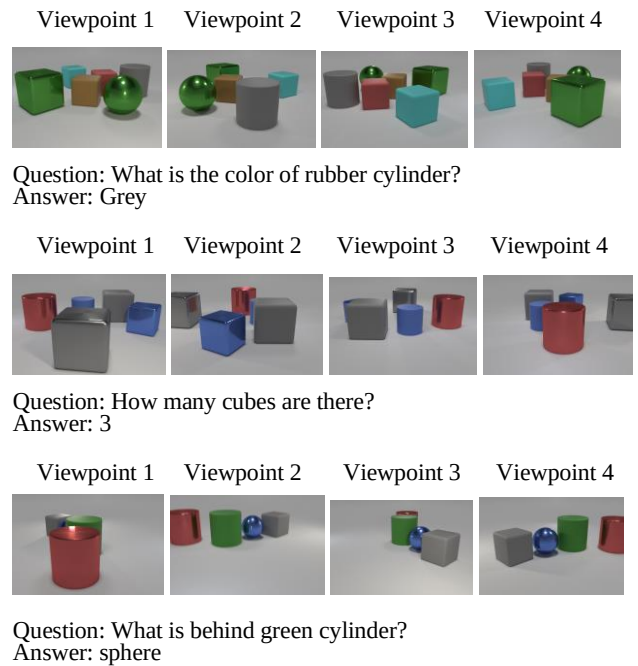


Figure 6: Examples of multi-view CLEVR dataset

V. RESULTS AND DISCUSSION

We first checked the accuracy and validity of our model by the means of f1 score and categorical cross entropy, below are the results for the same:

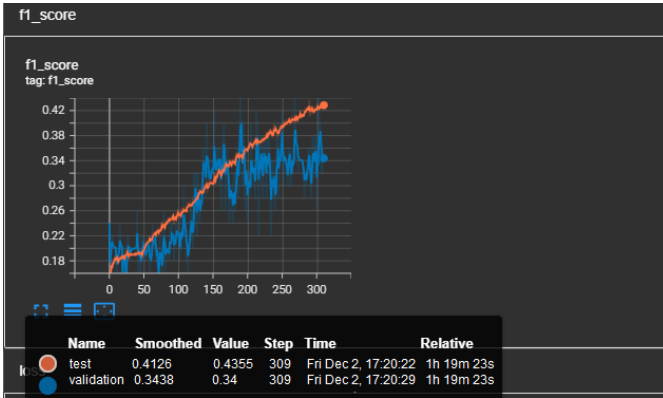


Figure 7: f1 score

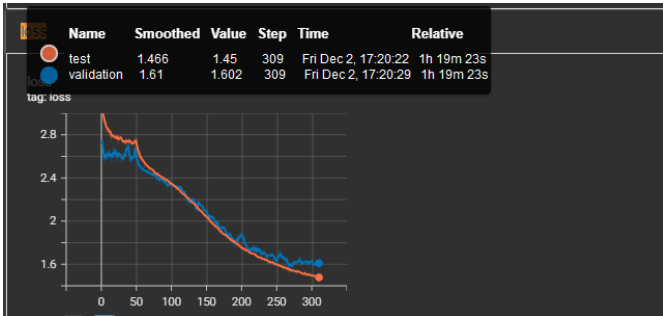


Figure 8: Categorical cross entropy loss

Talking about the results, if we check for the training portion the model is working fine with an accuracy of 56% and when it comes to the validation part where we are testing the actual functionality of 4 viewpoints it was giving an accuracy around 38 %.

VI. CONCLUSION

Our approach aims to improve the VQA task by utilizing 3D information obtained from multi-view images. While many recent VQA methods rely on a single image and question, this approach is limited by occlusions and insufficient visual information. To address this limitation, we propose integration 3D information from multi-view images directly into VQA, without relying on SLAM process. Our approach works on this challenging dataset, while also being practical for real-world application

We implemented Visual Question Answering on 3D Scenes using CLEVR dataset. For 4 view-point images we extracted image features using DenseNet121 [25] architecture of CNN and this we combined with text features which was extracted using LSTM using attention maps to get the final answer to the question. For training

we got 56% accuracy, for validation we got 43% accuracy and for testing we got 38% accuracy. The reason for getting lower accuracy on test set might be due to a smaller dataset as usually when it comes to visual question answering even on a single viewpoint it will have thousands of images and lakhs of questions and answers to train the model, hence this might be one of the reasons to get a lower accuracy.

REFERENCES

- [1] Damein Teney, Lingquiao Liu, Anton van den Hengel, July 2017, "Graph-Structured Representations for Visual Question Answering", 2017, IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
- [2] Qiu Yue, Ryota Suzuki, Hirokatsu Kataoka, Yutaka Satoh, September 2019, "Incorporation 3D Information Into Visual Question Answering", 2019 International Conference on 3D Vision (3DV).
- [3] Qingxing Cao, Xiaodan Liang, Bailin Li, Liang Lin, "Interpretable Visual question answering by reasoning on dependency trees", 2019, IEEE Transactions on Pattern Analysis and Machine Intelligence.
- [4] Mateusz Malinowski, Marcus Rohrbach, December 2015, "Ask Your Neurons: A Neural-based Approach to Answering Questions about Images", International conference on computer vision (ICCV).
- [5] Weidong Tian, Rencai Zhou, Zhong-Oiu Zhao, July 2020, "Cascading Top-Down Attention For Visual Question Answering", 2020 International Joint Conference on Neural Network (IJCNN).
- [6] Ryan Kiros, Ruslan Salakhutdinov, and Rich Zemel. Multimodal neural language models. In *International Conference on Machine Learning (ICML)*, pages 595–603, 2014.
- [7] Stanislaw Antol, Aishwarya Agrawal, Jiasen Lu, Margaret Mitchell, Dhruv Batra, C Lawrence Zitnick, and Devi Parikh. Vqa: Visual question answering. In *Proceedings of the IEEE international conference on computer vision (ICCV)*, 2015.
- [8] Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C Lawrence Zitnick, and Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual reasoning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [9] Akira Fukui, Dong Huk Park, Daylen Yang, Anna Rohrbach, Trevor Darrell, and Marcus Rohrbach. Multimodal compact bilinear pooling for visual question and visual grounding. In *Conference on Empirical Methods in Natural Language Processing*, 2016.
- [10] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision (ECCV)*, 2014.
- [11] Abhishek Das, Samyak Datta, Georgia Gkioxari, Stefan Lee, Devi Parikh, and Dhruv Batra. Embodied question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [12] Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [13] Hang Su, Subhransu Maji, Evangelos Kalogerakis, and Erik Learned-Miller. Multi-view convolutional neural networks for 3d shape recognition. In *Proceedings of the IEEE international conference on computer vision (CVPR)*, 2015.
- [14] Asako Kanezaki, Yasuyuki Matsushita, and Yoshifumi Nishida. Rotationnet: Joint object categorization and pose estimation using multiviews from unsupervised viewpoints. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [15] A. Allahyari and K. Borna, "Visual Question Answering on CLEVR Dataset via Multimodal Fusion and Relational Reasoning," *2021 52nd Annual Iranian Mathematics Conference (AIMC)*, Kerman, Iran, Islamic Republic of, 2021, pp. 74-76, doi: 10.1109/AIMC54250.2021.9656978.

- [16] J. Johnson, B. Hariharan, L. van der Maaten, L. Fei-Fei, C. L. Zitnick and R. Girshick, "CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning," *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017, pp. 1988-1997, doi: 10.1109/CVPR.2017.215.
- [17] R. Liu, C. Liu, Y. Bai and A. L. Yuille, "CLEVR-Ref+: Diagnosing Visual Reasoning With Referring Expressions," *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 2019, pp. 4180-4189, doi: 10.1109/CVPR.2019.00431.
- [18] J. Johnson *et al.*, "Inferring and Executing Programs for Visual Reasoning," *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017, pp. 3008-3017, doi: 10.1109/ICCV.2017.325.
- [19] C. Zhu, Y. Zhao, S. Huang, K. Tu and Y. Ma, "Structured Attentions for Visual Question Answering," *2017 IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017, pp. 1300-1309, doi: 10.1109/ICCV.2017.145.
- [20] S. Yang, X. Yu and Y. Zhou, "LSTM and GRU Neural Network Performance Comparison Study: Taking Yelp Review Dataset as an Example," *2020 International Workshop on Electronic Communication and Artificial Intelligence (IWECAL)*, Shanghai, China, 2020
- [21] T. Williams, D. Szafir, T. Chakraborti and E. Phillips, "Virtual, Augmented, and Mixed Reality for Human-Robot Interaction (VAM-HRI)," *2019 14th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Daegu, Korea (South), 2019, pp. 671-672, doi: 10.1109/HRI.2019.8673207.
- [22] S. Sarath and J. Amudha, "Visual question answering models Evaluation," *2020 International Conference for Emerging Technology (INCET)*, Belgaum, India, 2020, pp. 1-5, doi: 10.1109/INCET49848.2020.9154094.
- [23] X. Wang, S. Su, H. Huang and P. Chu, "Research on Mask RCNN based on rotating self-supervised learning," *2022 41st Chinese Control Conference (CCC)*, Hefei, China, 2022, pp. 6978-6981, doi: 10.23919/CCC55666.2022.9901579.
- [24] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [25] K. Zhang, Y. Guo, X. Wang, J. Yuan, Z. Ma and Z. Zhao, "Channel-Wise and Feature-Points Reweights Densenet for Image Classification," *2019 IEEE International Conference on Image Processing (ICIP)*, Taipei, Taiwan, 2019, pp. 410-414, doi: 10.1109/ICIP.2019.8802982.